

Amandla Construction EHS Dashboard

A free, deployable web app that replaces your manual Excel-based EHS reporting system.

What This Does (That Excel Doesn't)

Feature	Old Excel	This App
Add incidents	Type into 5+ sheets	One form → auto-updates everything
Calculate TIFR/LTIFR	Manual formulas	Auto-calculated live
Site data import	Copy-paste from site files	Upload → auto-extract
Monthly reports	Hours of copy-paste	One-click generate
View by division/site	Scroll through sheets	Click to filter instantly
Mobile access	None	Works on any phone
Charts	Static pasted images	Live, interactive

Files Included

File	Purpose
app.py	Main Streamlit application (35,000 lines)
requirements.txt	Python dependencies
README.md	This file

Deploy for FREE (3 Options)

Option 1: Streamlit Community Cloud (Easiest — Recommended)

1. **Create a GitHub account** (free) at github.com
2. **Create a new repository** called `amandla-ehs`
3. **Upload these 3 files** to the repository
4. **Go to** share.streamlit.io
5. **Sign in with GitHub**
6. **Click “New App”** → select your `amandla-ehs` repo
7. **Click Deploy** — done! You'll get a URL like `amandla-ehs.streamlit.app`

Cost: \$0 forever | **Time:** 5 minutes

Option 2: Render (Free, More Control)

1. **Push files to GitHub** (same as above)
2. **Go to** render.com
3. **Click “New Web Service”**
4. **Connect your GitHub repo**
5. **Set build command:** `pip install -r requirements.txt`
6. **Set start command:** `streamlit run app.py --server.port $PORT`
7. **Click Deploy**

Cost: \$0 (free tier) | **Time:** 10 minutes

Option 3: Run Locally (For Testing)

```
# 1. Install Python 3.9+ from python.org

# 2. Open terminal/command prompt

# 3. Create a folder
mkdir amandla-ehs
cd amandla-ehs

# 4. Save app.py and requirements.txt in this folder

# 5. Install dependencies
pip install -r requirements.txt

# 6. Run the app
streamlit run app.py

# 7. Open your browser to http://localhost:8501
```

How Sites Will Use It

Current Process (Painful):

1. Site fills Excel template → emails to you
2. You open email → download file → open master spreadsheet
3. Copy manhours into monthly sheet
4. Copy incidents into IOD table
5. Update all graph sheets
6. Recalculate TIFR/LTIFR manually
7. Fix broken formulas
8. Generate report → copy into Word/PDF

New Process (Simple):

1. Site opens app on phone/laptop
2. Clicks “ Import Site Data”
3. Uploads their monthly Excel file
4. App auto-extracts everything
5. You review and click “Approve”
6. All dashboards update instantly
7. Click “ Monthly Report” → download PDF/Excel

Next Steps to Make It Real

Phase 1: Pilot (This Week)

- Deploy to Streamlit Cloud
- Test with 1 site (e.g., D563 Muizenberg)
- Import your existing data
- Get feedback from site supervisors

Phase 2: Add Database (Month 1)

- Connect to SQLite/PostgreSQL (data persists)
- Add user logins (you = admin, sites = editors)
- Add audit trail (who changed what, when)

Phase 3: Full Production (Month 2-3)

- Mobile app (Progressive Web App)
- Offline capability for sites with poor signal
- Auto-email reports to management
- Integration with SharePoint (embed as web part)

💰 Cost Breakdown

Phase	Cost	What You Get
Now (Prototype)	R0	Full dashboard, incident register, import, reports
Month 1 (Database)	R0-R500/month	Persistent data, user accounts, audit trail
Month 3 (Full)	R0-R2,000/month	Mobile app, offline, auto-reports, SharePoint embed

Compare to commercial EHS software: **R8,000-15,000/month**

🆘 Support

If you get stuck deploying:

1. Check the [Streamlit deployment docs](#)
2. Message me with the error screenshot
3. Alternative: I can deploy it for you and send you the link

Data Privacy Note

- **Streamlit Cloud (free):** Data lives in memory while app runs. For persistent storage, add a database.
- **For sensitive data:** Use a private database (Supabase free tier, or your own server).
- **No data leaves your control** — you own the GitHub repo and the deployment.